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Model-driven Impact Quantification of Energy Resource Redundancy and Server Rejuvenation on Dependability of Medical Sensor Networks in Smart Hospitals

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Abstract: The Coronavirus (COVID-19) pandemic across countries all over the world urges governments to revolutionize the traditional medical hospitals/centers to provide sustainable and trustworthy medical services to patients under the pressure of huge overload on the computing systems of wireless sensor networks (WSN) for medical monitoring as well as treatment services of medical professionals. Uncertain malfunctions in any part of the medical computing infrastructure, from its power system in a remote area to the local computing systems at a smart hospital, can cause critical failures in medical monitoring services, which even lead to a fatal loss of human life in the worst case. Therefore, early design in the medical computing infrastructure's power and computing systems needs to carefully consider the dependability characteristics, including the reliability and availability of the WSN in smart hospitals under the uncertain outage of any part of the energy resources or failures of computing servers, especially due to software aging. In that regard, we propose in this paper reliability and availability models adopting stochastic Petri net (SPN) to quantify the impact of energy resources and server rejuvenation on the dependability of medical sensor networks. Three different availability models (A, B and C) are developed in accordance with various operational configurations of a smart hospital's computing infrastructure to assimilate the impact of energy resource redundancy and server rejuvenation techniques for high availability. Moreover, a comprehensive sensitivity analysis is performed to investigate the components that impose the greatest impact on the system availability. The analysis results indicate different impacts of the considered configurations on the WSN's operational availability in smart hospitals particularly, 99.40%, 99.53%, and 99.64% for the configurations A, B and C, respectively. This result highlights the difference of 21 hours per year in downtime when comparing the worst with the best case. This study can help leverage the early design of smart hospitals considering its wireless medical sensor networks' dependability quality of service to cope with the overloading of medical services in world-wide virus pandemics.

Keywords: Internet of things (IoT); smart hospital; energy resources; availability, stochastic Petri net

1. Introduction

The Internet of Medical Things (IoMT) has become an important computing paradigm in recent years due to the emergence of new diseases around the world in which wireless

30 medical devices and sensors are integrated and harmonized into a common network
31 in smart hospitals [1]. IoMT is a medical monitoring system that provides continuous
32 real-time monitoring and observation services to patients through wearable health sen-
33 sors and devices with wireless body area network (WBAN), artificial intelligence (AI),
34 and remote monitoring techniques [2]. With the functional advantages of IoMT, an early
35 warning system equipped with real-time data collection and storage can do in-depth
36 and rapid analysis to control the spread of infectious diseases and decrease the workload
37 of healthcare services. Wireless sensor data generated by hardware devices such as cell
38 phones are usually transmitted to a cloud/fog computing platform for decision making.
39 Therefore the operational continuity of the IoMT's overall infrastructure from its power
40 supply network in remote areas to the IT systems of local hospitals is essential in this
41 period.

42 Smart hospital infrastructures have many functions needed to keep them running
43 [3] such as (i) medical equipment (wireless sensors) for remote monitoring and remote
44 diagnostic diagnostics (ii) networked medical equipment (bracelet heart rate monitor
45 wireless thermometer, blood glucose meter, etc.) (iii) network device (transmission
46 media routers gateways, etc.) (iv) data (related to patient or staff information) and (v)
47 buildings and facilities (some sensors are distributed throughout the hospital). The
48 proper functioning and integration of these five macro-resources must be rigorously
49 ensured and maintained ongoing. Computer systems in hospitals often need to run as
50 efficiently as possible with minimal latency. However, local servers alone are not enough
51 to handle the large volumes of data generated during severe pandemic cases. Therefore,
52 using not only edge servers but also cloud servers is imperative. Such a complex and
53 distributed infrastructure involving sensors, actuators, and servers could not be missing.
54 In some cases, failure of one of the components can lead to a complete failure of the
55 patient monitoring system. Therefore, it is essential to carry out studies to ensure the
56 maximum availability of such systems.

57 Without exception, IoMT is certainly prone to partial crashes and system crashes
58 [4]. In the context of a pandemic, the ability to maintain 24/7 medical services from
59 your IoMT is essential and difficult due to the large number of data transactions that
60 can lead to unexpected events, unexpected medical events, and even serious losses
61 of life. The possibility of actual active incidents of IoMT in medical centers has been
62 demonstrated in reports in practice. At LDS Hospital in Salt Lake City, Utah, USA, a
63 computerized hospital information system called Health Assessment Through Logic
64 Programming (HELP) handles 17,000 logins per day [5,6]. A survey of Electronic Medical
65 Record (EMRD) downtime in a crowded urban emergency department from May 2016 to
66 December 2017 in [7] found there was a total of more *58 hours of downtime with 12 episodes*
67 *of EMRD* occurred during the study period with the interval *5 hours with unpredictable*
68 *intervals*. The EHR system at the National Institutes of Health Clinical Center (NIHCC) in
69 the United States unexpectedly closed on May 13, 2010, resulting in all patients suddenly
70 losing access to available clinical information, potentially affecting patient care and
71 safety [8]. These factors require IoMT researchers to understand the nature of availability
72 issues and their solutions through detailed modeling and system design evaluation even
73 before the system is deployed.

74 Assessing the availability of hospital computer systems is important but sometimes
75 unfeasible to perform in real environments. Hospital computer systems are sensitive
76 as they handle critical patient data. These systems and equipment usually have a high
77 monetary cost, making on-site experiments unfeasible. Two aspects that initially directly
78 impact the availability of smart hospitals are the electricity service quality provided to the
79 hospital and issues related to the software systems aging. If the hospital does not have
80 electricity, its systems will be off. If software performance is affected by computational
81 aging (memory leakage, for example), it will also cause general availability to drop
82 somewhat. Thus, the question that guides this paper is, therefore: *What is the impact of*

83 *using different energy sources and software rejuvenation methods under the availability of smart*
 84 *hospital computer systems?*

85 This work has two main contexts, energy issues regarding the smart hospital. The
 86 related work has focused on one of the two contexts in isolation. Oueida et al. [9] and
 87 Greco et al. [10] were the only works that focused on IoT. Araujo(b) et al. [11] was
 88 unique in the context of smart building. All experimental works in the energy context
 89 used reliability or availability metrics without directing them to the hospital context.
 90 Some smart hospitals performed availability analysis without observing energy issues
 91 and software aging. The works by Oueida et al. [9], Greco et al. [10], Chen et al. [12] and
 92 Araujo(a) et al. [13] used metrics more focused on the performance area. Therefore, to
 93 the best of our knowledge, our work is unique in exploring availability and reliability in
 94 the hospital context by looking at energy and software aging issues.

95 The availability analysis of hospital systems supported by IoMT is often unfeasible
 96 in real environments. Formal mathematical models such as queue networks, Markov
 97 chains, Petri nets can be adopted at an early design stage or evaluate complex configura-
 98 tions required to be represented in an operating system. This paper proposes a series
 99 of stochastic Petri net (SPN) models [14–16] to represent and evaluate a smart hospital
 100 architecture looking at energy issues and software aging. Petri net is a mathematical
 101 form based on probability theory that allows evaluating any system that undergoes
 102 some state change. SPN can represent synchronization, sequencing, parallelization,
 103 concurrency, among other aspects of any distributed system. SPN has already been
 104 adopted in previous studies in the hospital context, however, without focusing on energy
 105 issues and software aging [17,18]. Therefore, the main contributions of this paper are:

- 106 • **Three availability SPN models to evaluate the availability feature of a smart**
 107 **hospital system.** The models may calculate the availability of the IoMT system.
 108 Two models are the extensions of the first one. Thus, the first model includes only
 109 energy resources with a power grid and a diesel power generator that delivers
 110 energy to the hospital. The second model includes a redundant point at the energy
 111 supply by including a solar system energy resource. The third model explores
 112 a rejuvenation strategy under some important model components. These three
 113 models can be the reality of a real smart hospital, and this work can provide these
 114 models to guide system designers to optimize their infrastructures, for example.
- 115 • **A SPN model that calculates the IoMT reliability.** Reliability is the probability
 116 that the system has performed its function up to a predetermined and uninterrupted
 117 time limit. We have varied a specific parameter related to the cloud aging time in
 118 the third proposed configuration model. Thus experiments have shown that the
 119 cloud aging time impacts system reliability.
- 120 • **A sensitivity analysis of the base model.** The analysis has shown which compo-
 121 nents have the greatest impact on IoMT system availability. The power grid, for
 122 example, was the most impacting component in one of the IoMT system configura-
 123 tions. In other cases, the aging aspects of edge and cloud impacted more under the
 124 availability.

125 The remaining of this paper is organized as follows: Section 2 presents related
 126 works. Section 3 presents an overview of the architecture of the modeled system. Section
 127 4 presents the proposed SPN models, their functioning, and their peculiarities. Sec-
 128 tion 5 shows the results of the sensitivity analysis of the availability model. Section 6
 129 presents the results for three case studies, which serve as a practical guide for a system
 130 administrator. Finally, Section 7 concludes this work and discusses possible future work.

131

132 2. Related Work

133 This section presents some related works with similar approaches or contexts to
 134 this work. Oueida et al. [9] proposed a resource preservation net (RPN) framework
 135 using Petri nets. The work presents a framework capable of generating non-consumable

resource models that are theoretically described and validated. The work aims to measure some performance indicators of an intelligent hospital system with edge and cloud processing components. Among the performance metrics of the study, there is the patient's length of stay (LoS), resources usage rate, and average waiting time. Santos et al. [17] propose analytical models of Petri nets and Reliability Block Diagram (RBD) to assess the availability of an intelligent health monitoring system that depends on edge, fog, and cloud infrastructures. Santos et al. [17] still use a multi-objective optimization algorithm (NSGA-II) to improve system availability, taking into account its cost as a limitation.

Greco et al. [10] propose a technological and architectural solution based on Open Source big data technologies to perform real-time data flow analysis on wearable sensors. The architecture proposed by the work comprises four layers: the sensing layer, the pre-processing layer, the cluster processing layer, and the persistence layer. Each layer's performance analysis was performed to gauge each layer's memory and CPU usage. Chen et al. [12] proposed an Edge-Cognitive-Computing-based (ECC-based) smart-healthcare system. The system can monitor and analyze the status of patients using cognitive computing. Furthermore, the system can allocate resources according to the patient's degree of risk. Experiments have shown that the system improves the user experience, optimizes resources, and increases patient survival chances in sudden emergencies.

Araujo(a) et al. [13] proposes a high-level model capable of characterizing the behavior of a mHealth system. The objective of the work is to identify the probability of a system message being delivered in a t time. The paper does not analyze availability, but some parts of the model are characterized as an availability model. Lisboa et al. [19] propose a patient monitoring architecture using sensors and cloud and fog processing. The work also presents a sensitivity analysis that identifies the components that most impact system availability. Santos et al. [20] also propose a monitoring architecture using cloud and fog. However, [20] extends the idea of [19] and adds a model that can calculate performance metrics and identify possible bottlenecks in the system.

Rodrigues et al. [18] propose models capable of calculating performance and availability metrics in a smart hospitals system. The work presents a performance model capable of calculating Mean Response Time, Resource Utilization, and Discard. The work also presents an availability model and performs a sensitivity analysis on this model. The results show optimal settings for system performance and availability. The article by [18] is one of the two works that this work extends. The study by Araujo(b) et al. [11] proposes an energy availability model for intelligent construction. The paper investigates the impact of different types of solar panel energy systems on the availability of smart buildings. In addition, the work also makes a cost comparison for the adopted energy systems.

Diaz et al. [21] developed a methodological strategy to improve the conditions of autonomous photovoltaic systems through reliability research in the laboratory and rural areas. Collins et al. [22] propose an RBD model capable of calculating failure rates for different components of large photovoltaic systems. The work results show the relationship between availability and reliability concerning system uptime. The work by Sayed et al. [23] proposes an analysis of the availability, reliability, and maintainability of a photovoltaic system. RBD can represent myriad possibilities in terms of component relationships or parameters. Finally, Cai et al. [24] proposes a framework for evaluating the reliability of grid-connected photovoltaic systems with intermittent failures using Dynamic Bayesian Networks (DBN). The work uses the framework to assess the availability and reliability of different photovoltaic systems. Table 1 displays related works. The works were classified considering seven aspects: context, metrics, evaluation method, consider the use of energy, sensitivity analysis, availability, and reliability.

The first classification criterion is the work's **context**. The context concerns the main theme of the work. This work has two main contexts, energy issues, and smart hospitals.

Table 1: Related Works

Work	Context	Metrics	Evaluation Method	Consider the Use of Energy	Sensitivity Analysis	Availability	Reliability	Rejuvenation
[9]	Smart Healthcare and IoT	Patient Length of Stay (LoS), Resource Utilization, and Average Patient Waiting Time	Petri Net	No	No	No	No	No
[17]	Smart Healthcare and IoT	Availability	Stochastic Petri Net Layered Model	No	No	Yes	No	No
[10]	IoT	Resource Utilization	Practical Test	No	No	No	No	No
[12]	Smart Healthcare	QoS	Stochastic Petri Net	No	No	No	No	No
[13]	Smart Healthcare	Message Delivery Probability	Stochastic Petri Net	No	No	Yes	No	No
[19]	Smart Healthcare	Availability and Downtime	Stochastic Petri Net	No	Yes	Yes	No	No
[20]	Smart Healthcare	Availability, Throughput and Service Time	Stochastic Petri Net	No	Yes	Yes	No	No
[18]	Smart Healthcare	Mean Response Time, Resource Utilization, Discard, Availability and Downtime	Stochastic Petri Net	No	Yes	Yes	No	No
[11]	Smart Building and Energy	Availability and Energy Cost	Stochastic Petri Net	Yes	No	Yes	No	No
[21]	Energy	Reliability	Mathematical Model	Yes	No	No	Yes	No
[22]	Energy	Availability and Reliability	Reliability Block Diagram	Yes	No	Yes	Yes	No
[23]	Energy	Availability and Reliability	Reliability Block Diagram	Yes	No	Yes	Yes	No
[24]	Energy and Smart Healthcare	Availability and Reliability	Diagram Framework	Yes	No	Yes	Yes	No
This Work		Availability and Reliability	Stochastic Petri Net	Yes	Yes	Yes	Yes	Yes

The other works focused on one of the two contexts in isolation. Oueida et al. [9], Santos et al. [17] and Greco et al. [10] were the only works that focused on IoT. Araujo(b) et al. [11] was unique in the context of smart building. This work is focused on both the context of smart hospitals and the context of energy supply.

The second classification criterion was the investigated **metric**. All works in the energy context used reliability or availability metrics without directing them to the hospital context. Some smart hospitals performed availability analysis without observing energy issues and software aging. The works by Oueida et al. [9], Greco et al. [10], Chen et al. [12] and Araujo(a) et al. [13] used metrics more focused on the performance area. Our work is unique in exploring availability and reliability in the hospital setting by looking at energy issues and software aging.

The third-ranking criterion is the **evaluation method**. Analytical models were the most common method due to their practicality and agility in generating results. The most used analytical models were Petri nets and reliability block diagrams. Among the cited works, only Greco et al. [10] have chosen a layered model. Chen et al. [12] carried out a practical test. The work by Diaz et al. [21] was unique in that have adopted a low-level mathematical model to evaluate its project. Our work adopted the highly representative stochastic Petri net models.

The fourth classification criterion is to verify if the work **considered the energy use** in the architecture. Our work considers the use of energy for the hospital's internal functioning. The fifth classification criterion was the **sensitivity analysis**. Sensitivity analysis allows us to discover which components are more important to each system metric. The table shows that articles measuring availability often use this type of analysis. Availability is typically used in conjunction with sensitivity analysis as it points to the component that can be improved in the architecture. Our work uses sensitivity analysis to help with the availability and reliability metrics and to help with the component rejuvenation technique.

The sixth classification criterion was **availability**. Araujo(a) et al. [13] used the availability metric only to help measure the main metric, without exposing the results of the metric itself. The seventh criterion was **reliability**. The papers that performed a reliability analysis used it as the main metric with availability. Furthermore, again, the work by Araujo(a) et al. [13] considered the reliability to measure the main metric but did not show the reliability results. The last criterion was **rejuvenation**. This work was the only one surveyed that applied rejuvenation techniques while joining the context of smart hospitals with energy and verified their impact on system availability.

3. Architecture Overview

This section presents the characteristics of the modeled architecture. Figure 1 presents the IoMT architecture adopted in this paper. The architecture has been divided into two parts to make it easier to organize and understand. The Power System represents the entire power supply for the hospital. Smart Hospital represents the hospital itself and its internal components.

The Power System has three energy sources: Power Grid, Power Generator, and Solar Panel System. The Power Grid represents the public energy offered by public or private companies that, normally, have no connection with the hospital. The Power Generator is a power generator that normally runs on diesel, normally used in most hospitals for eventual power outages. The photovoltaic system consists of a solar panel, charger control, battery storage, solar inverter, and switch power. The Solar Panel represents one of the photovoltaic system sources that transform the thermal energy generated by the sun into electrical energy. Battery Storage is a batteries system that stores the produced solar energy. Batteries used to store solar energy are still considered limited nowadays [25]. Thus, batteries keep the hospital supplied for a short period until main power is restored. Charger Control regulates the amount of energy from the solar panel to the batteries. The Solar Inverter transforms the energy generated from

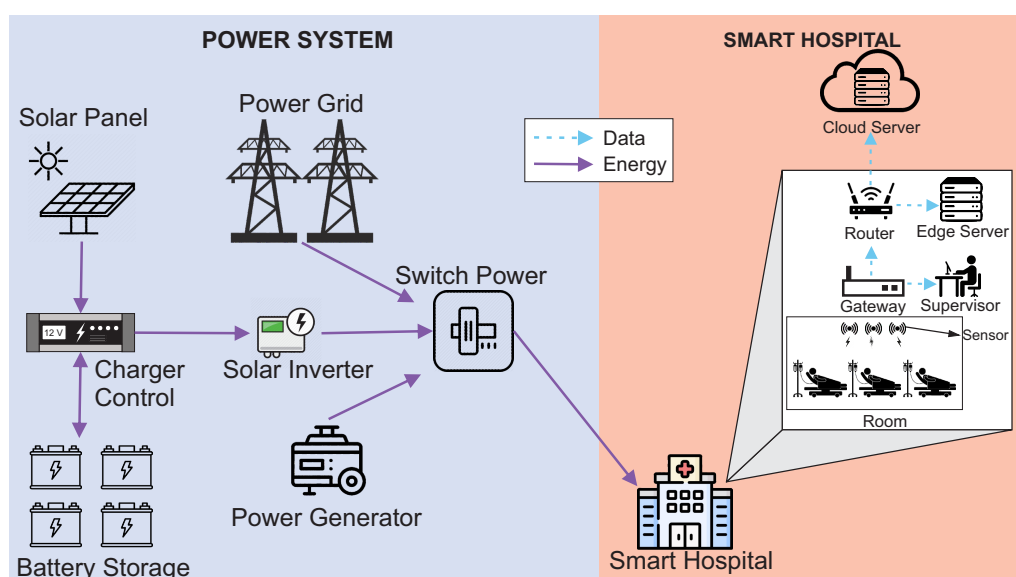


Figure 1. Proposed architecture of a smart hospital system.
Developed by authors.

243 direct current to alternating current. Switch Power is an actuator that decides which
244 energy source should be used: solar, public/private power grid, or generator. Finally,
245 the system works, considering that energy comes primarily from solar panels. Power
246 Grid will be used if solar power fails. The generator will be activated when the other
247 two resources fail.

248 Inside the Smart Hospital, some components will be used to monitor patients.
249 These components distribute information across the other components. The hospital
250 has several rooms with sensors that capture patient data from their beds. The sensors
251 are connected to the Gateway. The Gateway collects information from all sensors and
252 transmits it to the Router and Supervisor. The Supervisor is responsible for observing
253 and analyzing patient data in case of emergency. The Router will distribute the Gateway
254 data to an edge server located in the hospital and a remote cloud server. The Edge Server
255 will store the data locally at the hospital for reporting and any future queries. The Cloud
256 Server will store the data remotely to have a backup copy of the data, mainly to monitor
257 a patient remotely. Finally, all internal components of the hospital need the energy to be
258 available for as long as possible, and the Cloud Server is located at a remote place.

259 *Operational configurations:* To investigate the impact of energy resource redundancy
260 and server rejuvenation on the smart hospital's dependability, we consider three different
261 configurations as shown in Table 2. The configuration A represents a conventional smart
262 hospital, including a power grid and a generator. Configuration B represents a smart
263 hospital with a sustainable energy system. Configuration C reinforces the configuration
264 B against software aging problems running on cloud/edge servers. The configuration
265 C does not involve an additional physical component; instead, a server rejuvenation
266 mechanism at cloud and edge along with two logical components that control the
267 rejuvenation process is taken into account. The Rejuvenation Trigger is responsible for
268 managing the rejuvenation, while the Peakhour Indicator manages the rejuvenation
269 together with the Rejuvenation Trigger if there is a peak hour policy for the hospital. The
270 configuration B involves the components of the photovoltaic system, including the Solar
271 Panel, Battery System, Charger Control, Solar Inverter, and Switch Power.

272 4. Proposed SPN Models

273 In this section, availability and reliability SPN models are developed following the
274 considered operational configurations of the smart hospital's computing infrastructure.

Table 2: Operational configurations.

Config.	Smart Hospital	Power Grid	Generator	Photovoltaic System	Rejuvenation Trigger	Peakhour Indicator
A	X	X	X			
B	X	X	X	X		
C	X	X	X	X	X	X

4.1. Availability Models

This subsection presents the three SPN models focused on availability calculation.

4.1.1. Model A: Base architecture with basic power source

Figure 2 shows model A. Model A corresponds to configuration A presented in the last section. Therefore, Model A does not consider the photovoltaic system. This model represents a more common power source architecture for hospitals in general [26–28]. Model A will serve as a base model for comparison with the next models presented in this section. The hospital is an operative mode on Model A when all the internal components are active and when one of the external energy components is functioning. Two building blocks centralize the control of the two sides of the architecture using guard expressions, Power System, and Smart Hospital. Such control centralization enabled us to calculate the availability only focused on these two building blocks. All transitions with a red index mark are configured with a guard expression. Table 3 summarizes all guard expressions.

Table 3: Guard expressions for the model A transitions.

Guard Index	Guard Expression
g01	(#PG_U>0) OR (#EDG_U>0)
g02	(#PG_U<1) AND (#EDG_U<1)
g03	((#PS_U<1) OR (#IN_U<1) OR (#CC_U<1) OR ((#SP_U<1) AND (#BS_U<1))) AND (#PG_U<1)
g04	((#PS_U>0) AND (#IN_U>0) AND (#CC_U>0) AND ((#SP_U>0) OR (#BS_U>0))) OR (#PG_U>0)
g05	((#CS_U>0) AND (#ES_U>0) AND (#RT_U>0) AND (GT_U>0) AND (F_D=0) AND (SV_U>0)) AND (POWER_SYSTEM_U>0)
g06	(#CS_U<1) OR (#ES_U<1) OR (#RT_D>0) OR (GT_D>0) OR (F_D>0) OR (SV_D>0)
g07	(#POWER_SYSTEM_D>0)
g08	(#POWER_SYSTEM_U>0)
g09	(#POWER_SYSTEM_D>0)
g10	(#POWER_SYSTEM_U>0)
g11	(#POWER_SYSTEM_D>0)
g12	(#POWER_SYSTEM_U>0)
g13	(#POWER_SYSTEM_D>0)
g14	(#POWER_SYSTEM_U>0)
g15	(#POWER_SYSTEM_D>0)
g16	(#POWER_SYSTEM_U>0)
g20	(#ES_D>0)
g21	(#CS_D>0)

Figure 3 shows the Power System and Smart Hospital building block components in more detail. The Power System building block component represented the system's power status. Power System is considered up when there is a token at the place POWER_SYSTEM_U. Power System is not working when there is a token at the place POWER_SYSTEM_D. The change between active and inactive states is controlled by BLACKOUT and RESTORED transitions. BLACKOUT and RESTORED transitions have guard expressions. BLACKOUT transition is activated when all energy sources are inactive, given by expression $g01 = (\#PG_U > 0) \text{OR} (\#EDG_U > 0)$. RESTORED transition is activated when there is at least one power source active, given by expression $g02 = (\#PG_U < 1) \text{AND} (\#EDG_U < 1)$.

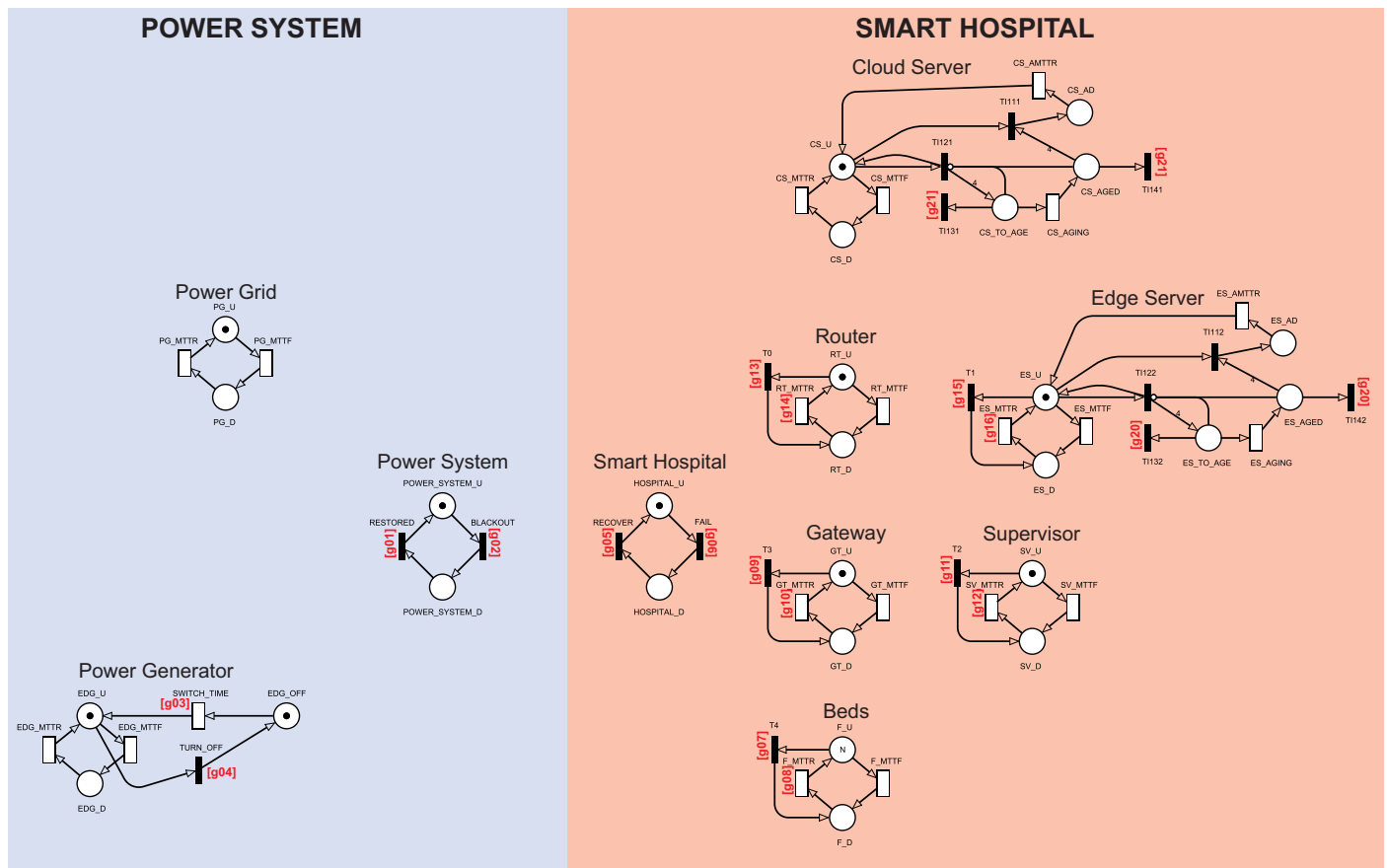


Figure 2. SPN model A.
Developed by authors.

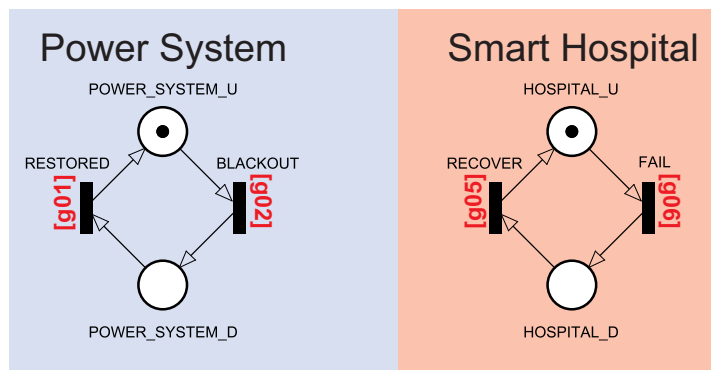


Figure 3. Power System and Smart Hospital building block control components.
Developed by authors.

299 The Smart Hospital building block component represents the hospital system status.
 300 The hospital is active when all the Smart Hospital components are active. The hospital is
 301 deactivated when at least one of the components has tokens in the inactive state. The
 302 Smart Hospital is up when there is a token in the place HOSPITAL_U. The Smart Hospital
 303 is not working when there is a token in the place HOSPITAL_D. The change between
 304 active and inactive states is caused by the transitions FAIL and RECOVER. Thus, FAIL
 305 transition is activated when all energy sources are inactive, given by expression $g05 =$
 306 $((\#CS_U > 0) \text{ AND } (\#ES_U > 0) \text{ AND } (\#RT_U > 0) \text{ AND } (GT_U > 0) \text{ AND } (F_D =$

0)AND(SV_U > 0))AND(POWER_SYSTEM_U > 0). RECOVER transition is activated when there is at least one energy source active, given by expression $g06 = (\#CS_U < 1)OR(\#ES_U < 1)OR(\#RT_D > 0)OR(GT_D > 0)OR(F_D > 0)OR(SV_D > 0)$.

Figure 4 shows the Power generator component more detailed. The power generator behaves differently from the other model components. The power generator has a cold standby mechanism [29]; that is, it will be only active when a certain condition is reached. The SWITCH_TIME transition will grant that the power generator will only be activated when there is no other active energy source. SWITCH_TIME is supported by the guard expression: $g03 = ((PS_U < 1)OR(\#IN_U < 1)OR(\#CC_U < 1)OR((\#SP_U < 1)AND(\#BS_U < 1)))AND(\#PG_U < 1)$. The TURN_OFF transition will grant that power generator will be immediately deactivated when any other energy source start working again, through expression $g04 = ((PS_U > 0)AND(\#IN_U > 0)AND(\#CC_U > 0)AND((\#SP_U > 0)OR(\#BS_U > 0)))OR(\#PG_U > 0)$

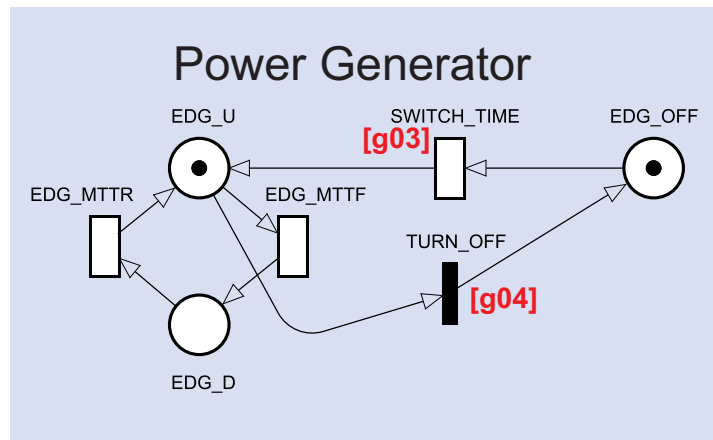


Figure 4. Power Generator component.
Developed by authors.

Figure 5 shows the aging mechanism that was used in detail. Software aging is modeled on Edge Server and Cloud Server through a four-phase Erlang subnet [30]. The number of phases depends on the parameters obtained for each component aging. The use of four aging phases is based on the previous study of Melo et al. [30]. The Erlang subnet can represent software aging by increasing the [31] failure rate. When the immediate transition TI122 triggers, four tokens (4 phases) move to the ES_TO_AGE place, and another token moves back to ES_U keeping the system alive and aging on the Erlang subnet. When the timed transition ES_AGING triggers, a token moves from ES_TO_AGE to ES_AGED representing the end of one of the aging phases. When ES_AGED has all four tokens, the immediate transition TI112 triggers, moving all tokens to place ES_AD, putting the system on aging failure.

The sum of the four phases of ES_AGING represents the total time for server aging. When the timed transition ES_AMTTR triggers, the system is repaired and becomes active again. The immediate transitions TI132 and TI142 “clean up” the aging, removing the tokens from ES_TO_AGE and ES_AGED. The aging cleansing process occurs when the system fails or starts the rejuvenation process. The rejuvenation process will be presented in Section 4.1.3. Immediate transition T1 shuts down Edge Server when Power System is unavailable due to failure. When T1 fires, a token moves from ES_U to ES_D, putting the Edge Server in a failed state. The same behaviors happen to the cloud server.

As the metric assumes that Power System and Smart Hospital building blocks must be in an active state for availability, different fault tolerance levels can be configured by changing their guard expressions. For example, the hospital can be considered active if at least one server or N of the M beds is active. We consider that all internal components must be active for the hospital to be available.

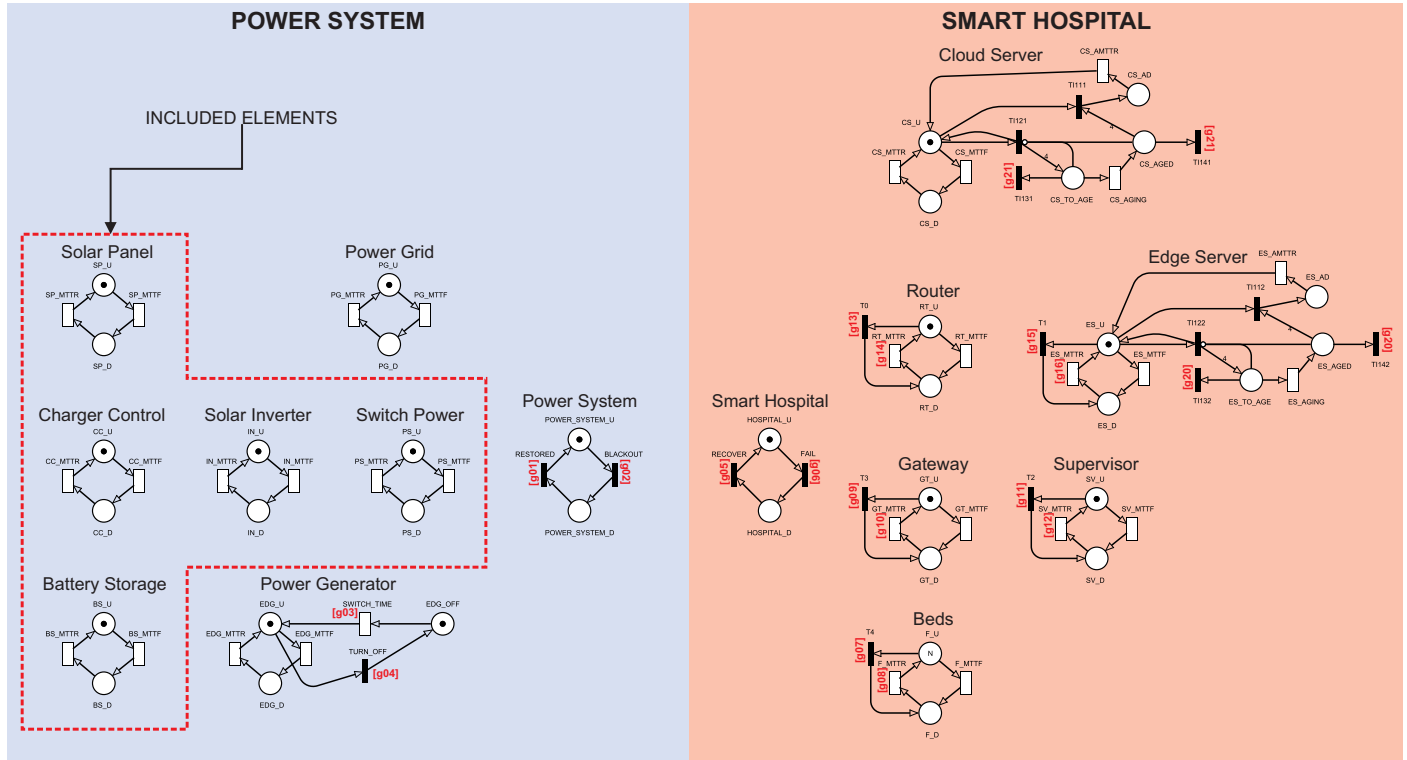


Figure 6. SPN Model B.
Developed by authors.

Equation 1, where A represents the system availability and 8760 the number of hours in the year. The equation that calculates availability for all models is given by the 2 Equation, that is, the probability that the Power System and Smart Hospital are both working at the same time. It is noteworthy that the user can change the guard expression of the central building blocks for other configurations.

$$D = (1 - A) \times 8760 \quad (1)$$

$$A = P\{(\#HOSPITAL_U > 0) \text{ AND } (\#POWER_SYSTEM_U > 0)\} \quad (2)$$

4.1.3. Model C: Model B with rejuvenation technique inclusion

Software rejuvenation is a technique that usually improves system availability [32]. Rejuvenation has the function of preventing crashes and reducing performance. The technique involves occasionally stopping the software from running to clear its state and restarting it [33]. Figure 7 shows model C that extends model B. The extension is based on the rejuvenation process under the cloud and edge servers. A Rejuvenation Trigger component represents peak times. The other energy and hospital components work the same as in Model B. Rejuvenation introduces planned interruptions. It can be advantageous to differentiate the peak and off-peak periods to plan the [33] rejuvenation strategy. We consider peak hours as day and off-peak hours as night periods, with 12 hours each. The peak component has two locations called PEAK_HOUR and OFFPEAK_HOUR. PEAK_HOUR and OFFPEAK_HOUR represent peak hour and off-peak hour, respectively. The peak component also has two timed transitions. When TIME_PEAK is triggered, it puts the system in peak hours. When TIME_OFFPEAK is triggered, the system goes to off-peak hours.

The sensitivity analysis presented later pointed out the Edge Server as the most sensitive component. Thus, the rejuvenation was applied to Edge Server since many

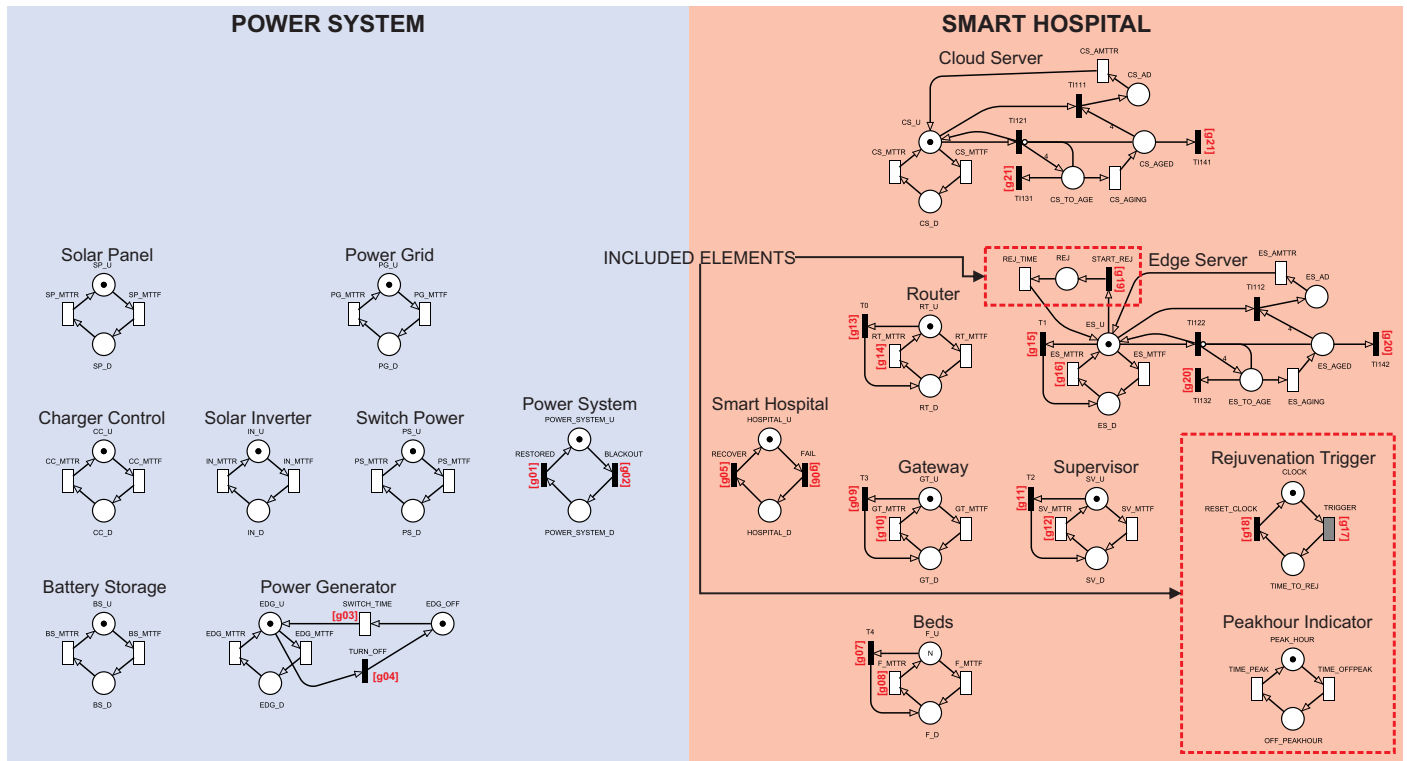


Figure 7. SPN Model C for availability.
Developed by authors.

studies prove that servers suffer from software aging and impacts on system availability [32,34,35]. The rejuvenation model was based on the model proposed by Garg et al. [36]. The place ES_U represents that the Edge Server is active. The place ES_D indicates that the Edge Server has failed and is disabled. When the ES_MTTF transition is triggered, a token moves from ES_U to ES_D, putting the Edge Server in a failed state. When the ES_MTR transition is triggered, the Edge Server is repaired, putting it back into the active state. The remaining component was explained in Section 4.1.1.

Another component presented in the C model is the Rejuvenation Trigger, which defines when the Edge Server will rejuvenate. The aging and rejuvenation process was based on the model proposed by [37]. The CLOCK place is used to store the Edge Server rejuvenation token. When the TRIGGER transition is triggered, a token moves from CLOCK to TIME_TO_REJ, indicating that the system is ready to begin rejuvenation. The TRIGGER represents the time-until-rejuvenate, that is, the time interval until the rejuvenation process occurs again. When TIME_TO_REJ has a token, the immediate transition START_REJ triggers, moving a token from ES_U to REJ, putting the system in a rejuvenation state. When REJ has a token, the immediate T0 transition triggers, moving the token from TIME_TO_REJ to CLOCK, starting the countdown for the next rejuvenation. When REJ has a token, the immediate transitions TI132 and TI142 clear the aging phase progress. When the timed transition REJ_TIME triggers, a token moves from REJ to ES_U, rejuvenating the system. The REJ_TIME transition represents the time to rejuvenate the Edge Server. During the rejuvenation process, the system is unavailable. Table 5 shows some guard transitions added and modified to grant the new behaviors of the rejuvenation mechanism.

The model uses two rejuvenation policies with peak times (no-peak policy and peak policy). The no-peak policy ignores peak times. The TRIGGER transition indicates the time for the rejuvenation to occur regardless of being in a peak period or not. The peak policy prevents the system starting rejuvenating during peak hours. When the

Table 5: New guard expressions for the model C transitions.

Guard Index	Guard Expression
g17*	(OFFPEAK_HOUR>0)
g18	(#REJ>0)
g19	(#TIME_TO_REJ>0)
g20	(#REJ>0) OR (#ES_D>0)

* only used if there is a peak hour policy.

system is off-peak (has a token in OFFPEAK_HOUR), the rejuvenation behaves the same as the no-peak policy. However, when the system is in peak hours (it has a token in PEAK_HOUR), the TRIGGER transition is not triggered even after the associated time unit has elapsed. Considering model C, the system is available when the Edge Server token is active (a token in ES_U. Finally, the equation 2 defines the availability of model C.

4.2. Reliability Model

Reliability is the conditional probability of a system remaining operational over a time interval $[0, t]$, considering that it was operational at $t = 0$. Figure 8 presents the reliability models of the proposed system. The three possible configurations were organized into a single figure. In principle, the model works the same as the availability SPN models. In this model, the components do not have the transitions that allow their recovery. The reliability of the presented models can be calculated using Equation 3. Finally, p calculates the probability that the system is unavailable. The equation can generate a curve that shows how reliability declines over time.

$$R = 1 - (P\{((\#HOSPITAL_D > 0)OR(\#POWER_SYSTEM_D > 0))\}) \quad (3)$$

5. Sensitivity Analysis

This section presents the sensitivity analysis. Table 6 shows the results for the sensitivity of the availability models components. The sensitivity experiment was obtained through stationary simulation. This decision was determined by the system's significant number of possible states. Due to many components, only the first ten results were selected from each model.

Table 6: Sensitivity analysis results for the 3 availability models.

Model A		Model B		Model C	
Component	Index	Component	Index	Component	Index
PG_MTTF	8.80×10^{-4}	ES_AGING	3.36×10^{-3}	CS_AGING	2.39×10^{-3}
ES_AGING	8.18×10^{-4}	CS_AGING	2.42×10^{-3}	ES_MTTF	1.90×10^{-3}
ES_MTTF	7.64×10^{-4}	ES_MTTR	2.37×10^{-3}	CS_MTTR	1.87×10^{-3}
ES_MTTR	7.53×10^{-4}	CS_MTTR	1.88×10^{-3}	CS_MTTF	1.51×10^{-3}
CS_AGING	6.06×10^{-4}	ES_MTTF	1.48×10^{-3}	ES_AGING	1.39×10^{-3}
CS_MTTR	4.38×10^{-4}	CS_MTTF	1.45×10^{-3}	ES_MTTR	1.00×10^{-3}
CS_MTTF	4.19×10^{-4}	GT_MTTF	6.31×10^{-5}	TIMECLOCK	1.05×10^{-4}
SW_MTTR	1.93×10^{-4}	SW_MTTF	5.73×10^{-5}	SW_MTTR	6.53×10^{-5}
GT_MTTR	1.59×10^{-4}	N	2.79×10^{-5}	SU_MTTF	6.24×10^{-5}
SE_MTTR	3.48×10^{-5}	CC_MTTF	2.20×10^{-5}	PG_MTTF	6.10×10^{-5}

Model A uses the Power Grid and a diesel generator to keep the hospital active. Thus, it can be seen that the Power Grid was the most critical component of the architecture; that is, its sensitivity index causes a significant impact on system availability. Power issues impact the final availability of the system, and Power Grid has a slightly more critical sensitivity index than the second place, which is edge aging time. From the second position on-wards (model A), all components are internal to the hospital, as the

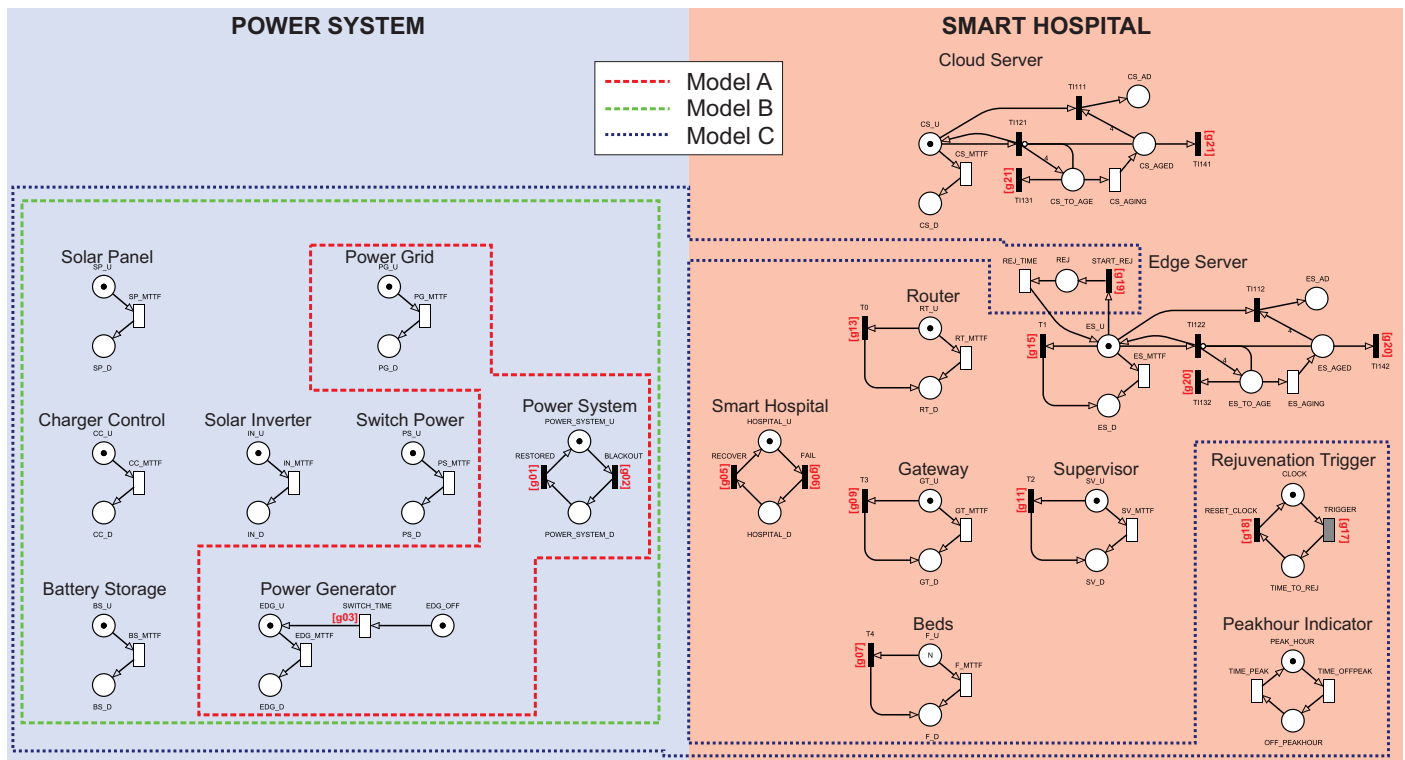


Figure 8. Reliability models created from availability models. Developed by authors.

generator is only occasionally turned on. The other two components that proved to be more critical are the edge and cloud servers due to their low failure and recovery times and their respective aging. Edge Server and Cloud Server can be considered critical because they presented a higher sensitivity index than the other components.

Model B brings redundancy when adding the photovoltaic system, an unusual configuration in hospitals today. The Power Grid index is no longer among the ten most critical. The edge aging (ES_AGING) sensitivity index becomes more critical and takes the first position, followed by cloud aging (CS_AGING). The photovoltaic system increases system availability. However, aging is still a critical factor for availability. The cloud and edge-related components were the most critical for model B, they were also more critical than most components in model A, but their values grew from 10^{-4} to 10^{-3} . The number of sensors represents a very small impact on the system, even though it appears among the ten most critical. The variation in the number of the sensor did not significantly impact availability.

In addition to proposing using the photovoltaic system, Model C also proposes rejuvenating the edge component. The edge server became more critical in Model B, so rejuvenation was applied. The most relevant indices for this model are still around 10^{-3} and decrease to values in 10^{-5} . The most critical index for this scenario is the cloud's aging time. The edge's aging index decreases due to the rejuvenation applied to the component. As the rejuvenation was not applied in the cloud, it remained with the same index. Edge and cloud failure and recovery components continue to occupy the top positions of the sensitivity table along with aging times. The criticality reduction of cloud and edge recovery and failure indices could be solved with some form of redundancy. However, the objective of this work was to check rejuvenation issues, which make the process cheaper.

Figure 6 shows in detail the variation of the two most critical indices for each model. Figure 9(a) shows the impact of PG_MTTF parameter variation on system availability.

Availability ranged from about 99.25% to 99.45%. As the curve presents an exponential behavior, it can be deduced that the less reliable the energy, the more abruptly the value tends to fall. Figure 9(b) shows the impact of edge aging in model A. The variation in the edge aging time caused the availability to vary greatly, but the maximum and average values were lower than those seen in the PG_MTTF variation. Figure 9(c) shows the impact of edge aging on model B availability.

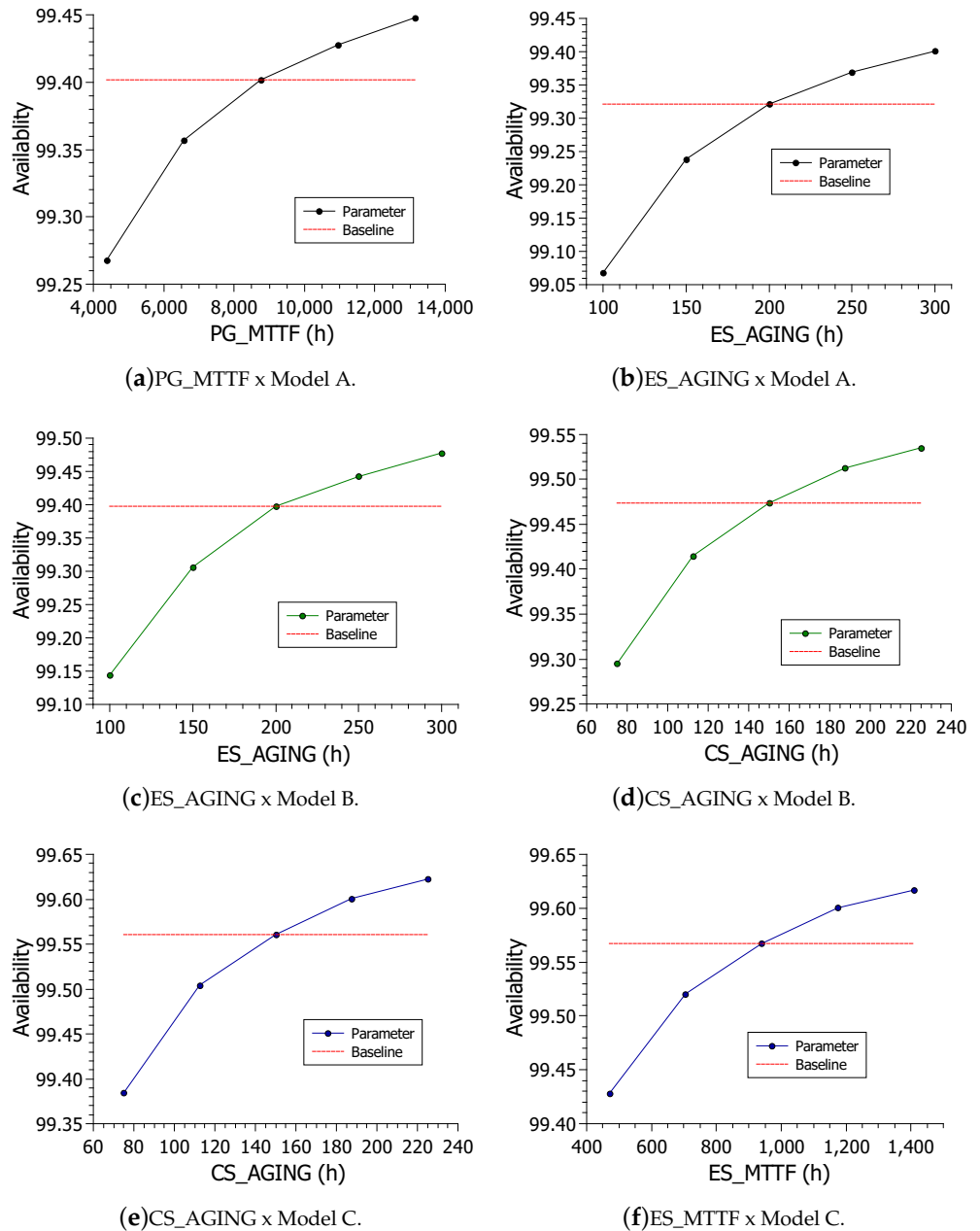


Figure 9. Results for the sensitivity of availability models.

Figure 9(d) shows the impact of cloud aging on model B. The cloud aging time variation resulted in slightly higher availability, but the edge aging had a large enough variation to surpass cloud aging time as the most critical component. Figure 9(e) shows the impact of the CS_AGING parameter on the availability of model C. Values range from about 99.38% to about 99.62%. The variation is still large, given the parameter's sensitivity index value, which is expected. However, the addition of edge rejuvenation caused the beginning and end of the sensitivity result interval to increase. The range in

the previous scenarios started just below 99.30%, whereas in this scenario, it starts above 99.30%. The same behavior is repeated for the upper range, increasing more than the model B sensitivity result. Finally, Figure 9(f) shows the impact of edge failure time on model C availability. Cloud aging had a noticeably longer range between values, so the edge recovery time was the second most important.

6. Case Studies

The results of different experiments with the models proposed in the previous section are presented. Table 7 presents the parameters used to feed the proposed models. The values used were taken from other validated studies. The values were taken from four papers [18], [11], [38] and [39]. The TIME_PEAK and TIME_OFFPEAK values represent daytime and nighttime respectively. The SWITCH_TIME has been set to 0.0833333 hours. Edge Server and Cloud Server aging times were based on their MTTFs, as for the work of [37]. The experiments were performed using the Mercury modeling tool [40]. All model transitions work in single server mode, except Beds component transitions. Beds component work in Infinite Server mode. Beds work in infinite server mode due to their nature, as the sensors work and are repaired in parallel. The availability experiment was obtained through stationary simulation due to the model size and rejuvenation particularities. The reliability experiment was carried out through transient simulation for the abovementioned reasons.

Table 7: Input parameters for proposed models.

Availability Parameters		
Component	MTTF (Hours)	MTTR (Hours)
Sensors/Actuators	300000	1
Gateway	480770	8
Supervisor	44957	1
Router	698220	8
Cloud Server	760	0.74
Edge Server	940	1.37
Solar Panel	219000	8
Battery System	47829	8
Charge Control	70080	8
Solar Inverter	24820	8
Emergency Diesel Generator	636	37
Power Grid	8757	4.807
Rejuvenation Parameters		
Transition	Time (Hours)	
TRIGGER	20-200 (steps of 20)	
REJ_TIME	0.0333333	
ES_AGING	200	
CS_AGING	150	
TIME_OFFPEAK	12	
TIME_PEAK	12	

6.1. Case study 1 - Availability and downtime analysis of the three models

In this section, the results of the availability model will be shown. Figure 10 shows the results for availability and downtime for different models. Figure 10(a) shows the results for availability. Model A had the lowest availability, with approximately 99.40% availability. With approximately 99.53% availability, Model B increased about 0.13% compared to model A. Model C achieved the highest available value, with approximately 99.64% availability. Overall, there was an increase of about 0.24% between models A and C. Model C showed an increase of 0.11% compared to model B proposed in this work.

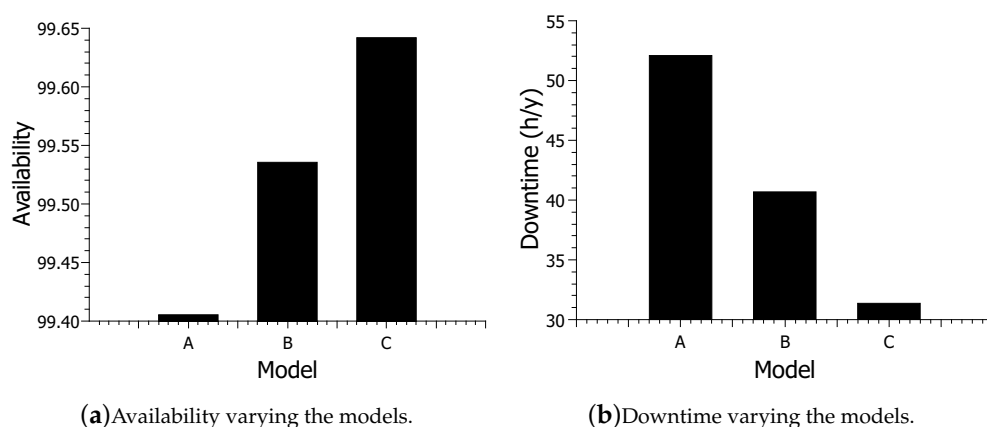


Figure 10. Results for availability models.

Figure 10(b) shows system downtime based on different Configurations. Model A showed the longest downtime, with approximately 52 hours of downtime. With approximately 40 hours of downtime, Model B showed a reduction of about 12 hours compared to model A. Model C had the lowest downtime, with around 31 hours of downtime. Model C shows a difference of about 21 hours concerning model A, almost a day of difference. The reduction in model C considering the B was about 9 hours.

The results for availability vary considerably little, but looking at the downtime results, we can see that it is an obvious difference. Availability has this slightly low variance, as power itself is a component that must be reliable; that is, it rarely fails. However, the inclusion of the solar energy system further increases the availability as it is a more reliable source of energy than the public one. The addition of the photovoltaic system showed the greatest increase in availability among the scenarios. Energy is indeed a component to be considered in modeling this type of system. Model C did not show an increase as big as model B, as it only adds a rejuvenating mechanism, but it is still a visible increase. Furthermore, a rejuvenation case in the cloud could increase the result of the last scenario. However, this paper has focused more on internal components in the hospital's control.

6.2. Case study 2 - Analysis of Availability and Downtime of Rejuvenation Policies in Model C

Figure 11 shows the availability and downtime of the rejuvenation model. Again, the simulation method was used with the Mercury tool [40] considering the availability model B and the two rejuvenation policies on model C. Figure 11(a) shows the availability of the system for model B and model C of rejuvenation in non-peak and peak policies. The deterministic transition time TRIGGER is varied in the simulations from 20 to 200, with intervals of 20 hours, based on the model proposed in [33]. When time-until-rejuvenate (TRIGGER) is at 20 hours, the no-peak policy has lower availability than the extended model, about 99.46% availability. After the time-until-rejuvenate reaches 40 hours, the no-peak policy availability improves significantly, outpacing the extended model availability.

The availability of the no-peak policy continues to increase as the time-until-rejuvenate increases. When the time until rejuvenate was at 20 hours, the rejuvenation happened so often that it hindered system availability. Too frequent rejuvenation impairs availability as the system goes down for a short period during rejuvenation. After the time-until-rejuvenate reaches 40 hours or more, rejuvenation occurs less frequently. Rejuvenation prevents the system from having more serious failures and improving availability. As time-until-rejuvenate increases, availability improves, as rejuvenation occurs at more strategic times until it peaks in 140 hours with an availability of 99.64%. After the time-until-rejuvenate exceeds 140 hours, rejuvenation begins to take too long

to occur, making the system prone to more serious failures and decreasing system availability.

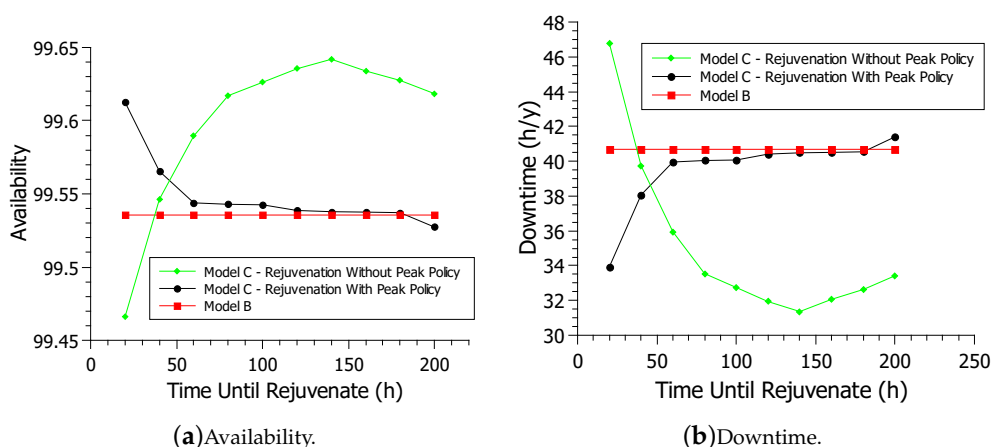


Figure 11. Comparison of levels of availability and downtime in relation to the rejuvenation model.

The peak policy behaves differently from the no-peak policy. When the time-until-rejuvenate is at 20 hours, the peak policy hits its highest availability peak at 99.61%. As time-until-rejuvenate increases, the availability of the no-peak policy decreases. The peak policy only rejuvenates if the system is off-peak, doing far fewer rejuvenations than the no-peak policy. The time-until-rejuvenate in 20 hours is already the ideal time for rejuvenating the peak policy. As the time-until-rejuvenate increases, rejuvenation takes a long time, impairing system availability. After exceeding 160 hours until rejuvenating, the peak policy availability is lower than that of the extended model. In general, rejuvenation was able to increase system availability greatly. All non-peak and peak policy values were better than model B values, especially shorter until rejuvenating. While the no-peak policy has far better availability than the peak policy, rejuvenating during peak hours may not be the best choice. The peak policy is also a smart choice in a scenario where this cannot happen, with better results than Model B, especially in 20 hours until rejuvenate.

Figure 11(b) shows the system downtime for model B and the rejuvenation model in both peak and non-peak hour policies. This graph reflects the availability graph, showing downtime over a year. When the time-until-rejuvenate is at 20 hours, the no-peak policy is at its best downtime of 33.92 hours. As time-until-rejuvenate increases, downtime also increases. The lower the availability, the greater the downtime. After the time-until-rejuvenate passes 180 hours, the peak policy's downtime reaches 41.30 hours, surpassing the 40.68 hours of model B. The no-peak policy has the longest downtime at 46.78 hours when the time until rejuvenating is 20 hours. However, downtime decreases as the time-until-rejuvenate increases, until 31.36 hours per year when the time-until-rejuvenate is 140 hours. After 140 hours until rejuvenate, the downtime starts to increase, reaching 33.42 hours when the time-until-rejuvenate is 200 hours. Therefore, the two rejuvenation policies had a lower downtime than model B, especially when each policy's time-until-rejuvenate is at its peak.

6.3. Case study 3 - Analysis of the Impact of Time Until Rejuvenate and Edge MTTF on Availability

Figure 12 presents a 3D surface graph to show the system's behavior considering availability by varying two factors with a high impact on availability. Edge MTTF and Time Until Rejuvenate factors were varied in 5 ranges: base value, minus 25%, minus 50%, plus 25%, and 50%. Colors are related to the result of availability. The bar on the right indicates the magnitude of the results. The upper part of the bar

indicates the highest availabilities, while the lower part indicates the lowest availabilities achieved. So the red color means the highest availability, and the purple color means the lowest availability. Changing the MTTF is more impactful than changing the Time Until Rejuvenate. The color red is present in most of the Edge MTTF projections. If the MTTF value is low, the availability always tends to be low, even changing the Time Until Rejuvenate. Therefore, the result indicates that increasing the MTTF improves system availability. However, finding an ideal value for Time Until Rejuvenate will also benefit the system's availability.

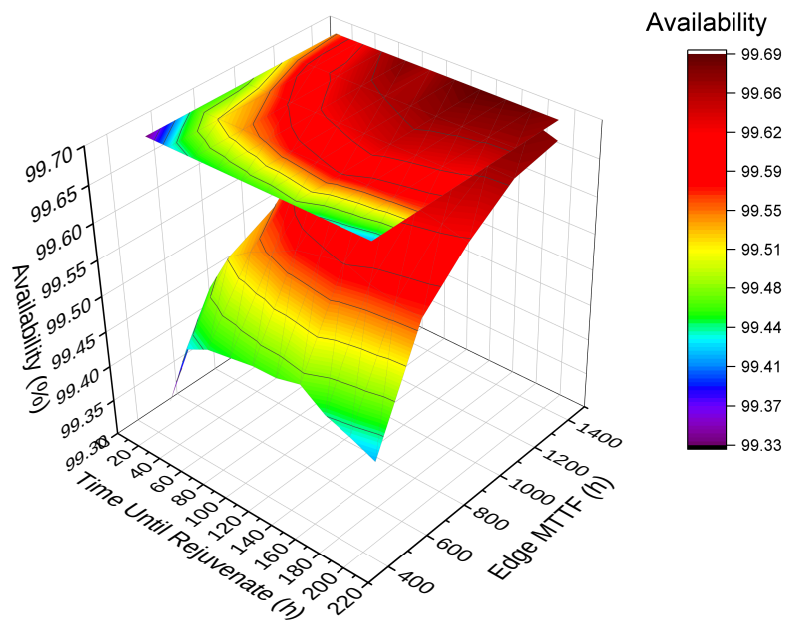


Figure 12. Availability analysis by varying concomitant factors.

6.4. Case study 4 - Reliability Analysis

This section displays the results for the model's reliability metric. Figure 13 shows the results for reliability over 20000 hours varying the cloud aging time (CS_AGING) for the C model. The cloud aging time was chosen due to its high sensitivity index. There are three reliability models. However, varying model C proved more relevant when analyzing the sensitivity analysis results. The cloud does not have a rejuvenation mechanism. Thus, the cloud became the most critical component system. The experiment simulates a reliability analysis in scenarios where the cloud cannot be rejuvenated, and a more or less reliable cloud must be chosen. All results decrease over time, as this is the nature of this experiment. In other words, the reliability of a system always tends to decrease over time.

The scenario where the cloud has aging time reduced by 50% presented values that remain below the values presented in the other two scenarios. The base scenario also touches the X-axis at values above 20000 hours. The scenario where the cloud has its aging time increased by 50%, has shown better results than all other scenarios, staying above the others from the start and following through for practically the entire time interval.

The +50% scenario also touches the X-axis at values greater than the 20000 hours of the experiment as in the other scenarios. The +50% scenario remains above the other scenarios at the 20,000-hour endpoint on the chart, with around 3% of confidence, versus around 2% for the other two scenarios. The results show a trend towards increased reliability in obtaining a slower aging cloud, which is expected. The main point of this

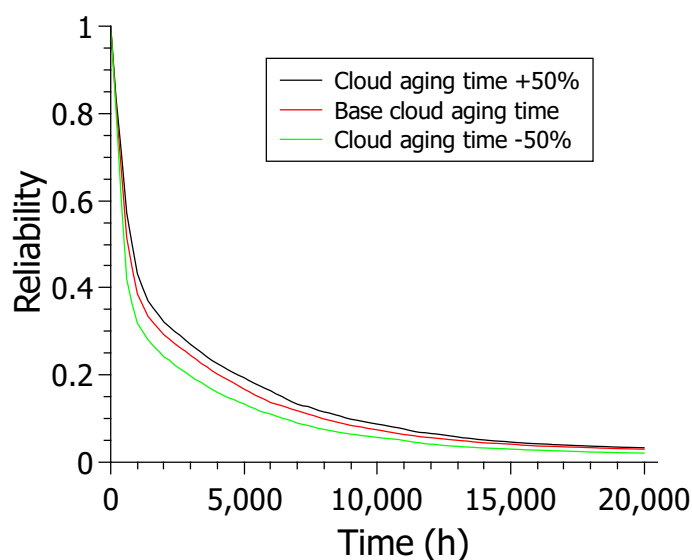


Figure 13. Reliability results varying the CS_AGING parameter in model C.

case study is that, given a budget, it is feasible to select the most advantageous cloud for the company as long as it meets the established dependability criteria. For example, if there is a need for a cloud that makes the system at least 20% reliable after 3000 hours, the base cloud would be enough to meet the demand without investing more in a more expensive cloud.

7. Conclusions and Future Works

This paper proposed stochastic Petri net models for an architecture of intelligent hospitals to help system administrators plan computing architectures. The models consider several factors that influence the total availability of the system. Energy is one of the big factors that were considered, and the use of solar energy, in addition to making the hospital more sustainable, showed a considerable increase in availability. The aging of more complex components such as servers has also had a major impact on system availability. The use of rejuvenation to treat aging was considered in the model, and the analysis shows that the model can estimate the ideal time between rejuvenation for a given model. The availability model can be configured using about 20 different parameters, while the reliability model has about half that amount.

Models provide accurate estimates of availability, downtime, and reliability metrics. The models were demonstrated by carrying out 4 case studies, 1 of them for the reliability analysis and the other three for further availability analysis. The results show how each model behaves by varying parameters by sensitivity analysis. The case studies provide a practical guide that shows how a system administrator can apply the model to perform assessments of various configurations for a sustainable smart hospital architecture.

This work presents some limitations that are important to highlight: (i) Aiming to avoid the “state-space explosion” problem, a couple of simplifications were necessary for the models. There are complex components such as power grid and cloud server, for example, that are treated as encapsulated components with single corresponding MTTF/MTTR parameters. (ii) availability is sometimes influenced by user interaction, security risks, and environmental issues. None of these aspects were investigated, only aging processes. (iii) the rejuvenation strategy was applied only at cloud/edge servers. However, the other hardware equipment (router, supervisor, and gateway) may also suffer from aging. We did not consider that. (iv) we have used only heterogeneous redundancy (distinct power supply sources) at the power system side. The hospital can have, for example, more than one power diesel generator, but we did not consider such a possibility. Future work intends to carry out a performance analysis to verify the impact

that the availability of components can have on the response time and throughput of the system. More external factors can also be considered, such as the functioning of the photovoltaic system on cloudy days.

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References

1. Nguyen, T.A.; Fe, I.; Brito, C.; Kaliappan, V.K.; Choi, E.; Min, D.; Lee, J.W.; Silva, F.A. Performability Evaluation of Load Balancing and Fail-over Strategies for Medical Information Systems with Edge/Fog Computing Using Stochastic Reward Nets. *Sensors* **2021**, *21*, 6253.
2. Santos, L.; Cunha, B.; Fé, I.; Vieira, M.; Silva, F.A. Data Processing on Edge and Cloud: A Performability Evaluation and Sensitivity Analysis. *Journal of Network and Systems Management* **2021**, *29*, 1–24.
3. Said, A.M.; Yahyaoui, A.; Abdellatif, T. Efficient Anomaly Detection for Smart Hospital IoT Systems. *Sensors* **2021**, *21*, 1026.
4. Zou, X.; Dai, Y.S.; Doebbeling, B.; Qi, M. Dependability and security in medical information system. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 2007, Vol. 4553 LNCS, pp. 549–558. doi: 10.1007/978-3-540-73111-5_63.
5. Gardner, R.M.; Allan Pryor, T.; Warner, H.R. The HELP hospital information system: Update 1998. *International Journal of Medical Informatics* **1999**, *54*, 169–182. doi:10.1016/S1386-5056(99)00013-1.
6. Nelson, N.C. Downtime procedures for a clinical information system: a critical issue. *Journal of Critical Care* **2007**, *22*, 45–50. doi:10.1016/j.jcrc.2007.01.004.
7. Jenkins, D.; Qureshi, R.S.; Moinudheen, J.; Pathan, S.A.; Thomas, S.H. Evaluation of electronic medical record downtime in a busy emergency department. *Qatar Medical Journal* **2020**, *2020*. doi:10.5339/QMJ.2020.20.
8. Patricia S. Coffey, RHIA, C.; Susan Postal, DNP, R.B.; Susan M. Houston, RN-BC, PMP, CPHIMS, F.; Jon W. McKeeby, DSc, MBA, C. Lessons Learned from an Electronic Health Record Downtime | Perspectives, 2016.
9. Oueida, S.; Kotb, Y.; Aloqaily, M.; Jararweh, Y.; Baker, T. An edge computing based smart healthcare framework for resource management. *Sensors* **2018**, *18*, 4307.
10. Greco, L.; Ritrovato, P.; Xhafa, F. An edge-stream computing infrastructure for real-time analysis of wearable sensors data. *Future Generation Computer Systems* **2019**, *93*, 515–528.
11. Araujo, E.; Pereira, P.; Dantas, J.; Maciel, P. Dependability Impact in the Smart Solar Power Systems: An Analysis of Smart Buildings. *Energies* **2021**, *14*, 124.
12. Chen, M.; Li, W.; Hao, Y.; Qian, Y.; Humar, I. Edge cognitive computing based smart healthcare system. *Future Generation Computer Systems* **2018**, *86*, 403–411.
13. Araujo, J.; Silva, B.; Oliveira, D.; Maciel, P. Dependability evaluation of a mhealth system using a mobile cloud infrastructure. 2014 IEEE International Conference on Systems, Man, and Cybernetics (SMC). IEEE, 2014, pp. 1348–1353.
14. Carvalho, D.; Rodrigues, L.; Endo, P.T.; Kosta, S.; Silva, F.A. Mobile Edge Computing Performance Evaluation using Stochastic Petri Nets. 2020 IEEE Symposium on Computers and Communications (ISCC). IEEE, 2020, pp. 1–6.

- 708 15. Carvalho, D.; Rodrigues, L.; Endo, P.T.; Kosta, S.; Silva, F.A. Edge servers placement in
709 mobile edge computing using stochastic Petri nets. *International Journal of Computational*
710 *Science and Engineering* **2020**, *23*, 352–366.
- 711 16. Silva, F.A.; Rodrigues, M.; Maciel, P.; Kosta, S.; Mei, A. Planning mobile cloud infrastructures
712 using stochastic petri nets and graphic processing units. 2015 IEEE 7th International
713 Conference on Cloud Computing Technology and Science (CloudCom). IEEE, 2015, pp.
714 471–474.
- 715 17. Santos, G.L.; Gomes, D.; Kelner, J.; Sadok, D.; Silva, F.A.; Endo, P.T.; Lynn, T. The inter-
716 net of things for healthcare: optimising e-health system availability in the fog and cloud.
717 *International Journal of Computational Science and Engineering* **2020**, *21*, 615–628.
- 718 18. Rodrigues, L.; Gonçalves, I.; Fé, I.; Endo, P.T.; Silva, F.A. Performance and availability
719 evaluation of an smart hospital architecture. *Computing* **2021**, *103*, 2401–2435.
- 720 19. da Silva Lisboa, M.F.F.; Santos, G.L.; Lynn, T.; Sadok, D.; Kelner, J.; Endo, P.T.; others.
721 Modeling the availability of an e-health system integrated with edge, fog and cloud infras-
722 tructures. 2018 IEEE Symposium on Computers and Communications (ISCC). IEEE, 2018,
723 pp. 00416–00421.
- 724 20. Santos, G.L.; Endo, P.T.; da Silva Lisboa, M.F.F.; da Silva, L.G.F.; Sadok, D.; Kelner, J.; Lynn,
725 T.; others. Analyzing the availability and performance of an e-health system integrated with
726 edge, fog and cloud infrastructures. *Journal of Cloud Computing* **2018**, *7*, 1–22.
- 727 21. Diaz, P.; Egido, M.A.; Nieuwenhout, F. Dependability analysis of stand-alone photovoltaic
728 systems. *Progress in photovoltaics: Research and applications* **2007**, *15*, 245–264.
- 729 22. Collins, E.; Dvorack, M.; Mahn, J.; Mundt, M.; Quintana, M. Reliability and availability
730 analysis of a fielded photovoltaic system. 2009 34th IEEE Photovoltaic Specialists Conference
731 (PVSC). IEEE, 2009, pp. 002316–002321.
- 732 23. Sayed, A.; El-Shimy, M.; El-Metwally, M.; Elshahed, M. Reliability, availability and maintain-
733 ability analysis for grid-connected solar photovoltaic systems. *Energies* **2019**, *12*, 1213.
- 734 24. Cai, B.; Liu, Y.; Ma, Y.; Huang, L.; Liu, Z. A framework for the reliability evaluation of
735 grid-connected photovoltaic systems in the presence of intermittent faults. *Energy* **2015**,
736 *93*, 1308–1320.
- 737 25. Hassan, M.; Saha, S.; Haque, M.E. A framework for the performance evaluation of household
738 rooftop solar battery systems. *International Journal of Electrical Power & Energy Systems* **2021**,
739 *125*, 106446.
- 740 26. Kolokotsa, D.; Tsoutsos, T.; Papantoniou, S. Energy conservation techniques for hospital
741 buildings. *Advances in Building Energy Research* **2012**, *6*, 159–172, [<https://doi.org/10.1080/17512549.2012.672007>].
742 doi:10.1080/17512549.2012.672007.
- 743 27. Yoshida, S.; Ito, K.; Yokoyama, R. Sensitivity analysis in structure optimization of energy
744 supply systems for a hospital. *Energy Conversion and Management* **2007**, *48*, 2836–2843. 19th
745 International Conference on Efficiency, Cost, Optimization, Simulation and Environmental
746 Impact of Energy Systems, doi:<https://doi.org/10.1016/j.enconman.2007.06.045>.
- 747 28. Ji, R.; Qu, S. Investigation and Evaluation of Energy Consumption Performance for Hospital
748 Buildings in China. *Sustainability* **2019**, *11*. doi:10.3390/su11061724.
- 749 29. Araújo, G.; Rodrigues, L.; Oliveira, K.; Fé, I.; Khan, R.; Silva, F.A. Vehicular cloud computing
750 networks: availability modelling and sensitivity analysis. *International Journal of Sensor*
751 *Networks* **2021**, *36*, 125–138.
- 752 30. Melo, M.; Maciel, P.; Araujo, J.; Matos, R.; Araujo, C. Availability study on cloud computing
753 environments: Live migration as a rejuvenation mechanism. 2013 43rd annual IEEE/IFIP
754 international conference on Dependable systems and networks (DSN). IEEE, 2013, pp. 1–6.
- 755 31. Machida, F.; Kim, D.S.; Trivedi, K.S. Modeling and analysis of software rejuvenation in a
756 server virtualized system with live VM migration. *Performance Evaluation* **2013**, *70*, 212–230.
- 757 32. Matos, R.; Araujo, J.; Alves, V.; Maciel, P. Experimental evaluation of software aging effects
758 in the eucalyptus elastic block storage. 2012 IEEE International Conference on Systems, Man,
759 and Cybernetics (SMC). IEEE, 2012, pp. 1103–1108.
- 760 33. Wang, D.; Xie, W.; Trivedi, K.S. Performability analysis of clustered systems with rejuvenation
761 under varying workload. *Performance Evaluation* **2007**, *64*, 247–265.
- 762 34. Huang, Y.; Kintala, C.; Kolettis, N.; Fulton, N.D. Software rejuvenation: Analysis, module
763 and applications. Twenty-fifth international symposium on fault-tolerant computing. Digest
764 of papers. IEEE, 1995, pp. 381–390.

-
- 765 35. Torquato, M.; Araujo, J.; Umesh, I.; Maciel, P. Sware: a methodology for software aging and
766 rejuvenation experiments. *Journal of Information Systems Engineering and Management* **2018**,
767 3, 15.
- 768 36. Garg, S.; Puliafito, A.; Telek, M.; Trivedi, K.S. Analysis of software rejuvenation using Markov
769 regenerative stochastic Petri net. Proceedings of Sixth International Symposium on Software
770 Reliability Engineering. ISSRE'95. IEEE, 1995, pp. 180–187.
- 771 37. Torquato, M.; Umesh, I.; Maciel, P. Models for availability and power consumption evaluation
772 of a private cloud with VMM rejuvenation enabled by VM Live Migration. *The Journal of*
773 *Supercomputing* **2018**, 74, 4817–4841.
- 774 38. Marqusee, J.; Ericson, S.; Jenket, D. Emergency Diesel Generator Reliability and Installation
775 Energy Security. Technical report, National Renewable Energy Lab.(NREL), Golden, CO
776 (United States), 2020.
- 777 39. Perdue, M.; Gottschalg, R. Energy yields of small grid connected photovoltaic system: effects
778 of component reliability and maintenance. *IET Renewable Power Generation* **2015**, 9, 432–437.
- 779 40. Silva, B.; Matos, R.; Callou, G.; Figueiredo, J.; Oliveira, D.; Ferreira, J.; Dantas, J.; Lobo, A.;
780 Alves, V.; Maciel, P. Mercury: An integrated environment for performance and dependability
781 evaluation of general systems. Proceedings of Industrial Track at 45th Dependable Systems
782 and Networks Conference, DSN, 2015.

